

Introduction to Electronic Design Automation

Homework 3

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1 BDD Operation

1. (a)

$$\begin{aligned}
 & \text{robdd_build}(ab(\neg c + d) + (a\neg b + \neg ab)(c\neg d + \neg cd), 1) \\
 & \xrightarrow{\eta} \text{robdd_build}(b(\neg c + d) + \neg b(c\neg d + \neg cd), 2) \\
 & \xrightarrow{\eta} \text{robdd_build}(\neg c + d, 3) \\
 & \xrightarrow{\eta} \text{robdd_build}(d, 4) \\
 & \xrightarrow{\eta} \text{robdd_build}(1, 5) \\
 & \quad v_1 \\
 & \xrightarrow{\lambda} \text{robdd_build}(0, 5) \\
 & \quad v_0 \\
 & \quad v_2 = (d, v_1, v_0) \\
 & \xrightarrow{\lambda} \text{robdd_build}(1, 4) \\
 & \quad v_1 \\
 & \quad v_3 = (c, v_2, v_1) \\
 & \xrightarrow{\lambda} \text{robdd_build}(c\neg d + \neg cd, 3) \\
 & \xrightarrow{\eta} \text{robdd_build}(\neg d, 4) \\
 & \xrightarrow{\eta} \text{robdd_build}(0, 5) \\
 & \quad v_0 \\
 & \xrightarrow{\lambda} \text{robdd_build}(1, 5) \\
 & \quad v_1 \\
 & \quad v_5 = (d, v_0, v_1) \\
 & \xrightarrow{\lambda} \text{robdd_build}(d, 4) \\
 & \quad v_2 = (d, v_1, v_0) \\
 & \quad v_4 = (c, v_5, v_2) \\
 & \quad v_6 = (b, v_3, v_4) \\
 & \xrightarrow{\lambda} \text{robdd_build}(b(c\neg d + \neg cd), 2) \\
 & \xrightarrow{\eta} \text{robdd_build}(c\neg d + \neg cd, 3) \\
 & \quad v_4 = (c, v_5, v_2) \\
 & \xrightarrow{\lambda} \text{robdd_build}(0, 3) \\
 & \quad v_0 \\
 & \quad v_7 = (b, v_4, v_0) \\
 & \quad v_8 = (a, v_6, v_7)
 \end{aligned}$$

1. (b)

Since

$$f = ab(\neg c + d) + (a\neg b + \neg ab)(c\neg d + \neg cd)$$

we have

$$f_c = abd + (a\neg b + \neg ab)(\neg d)$$

$$f_{\neg c} = ab + (a\neg b + \neg ab)(d)$$

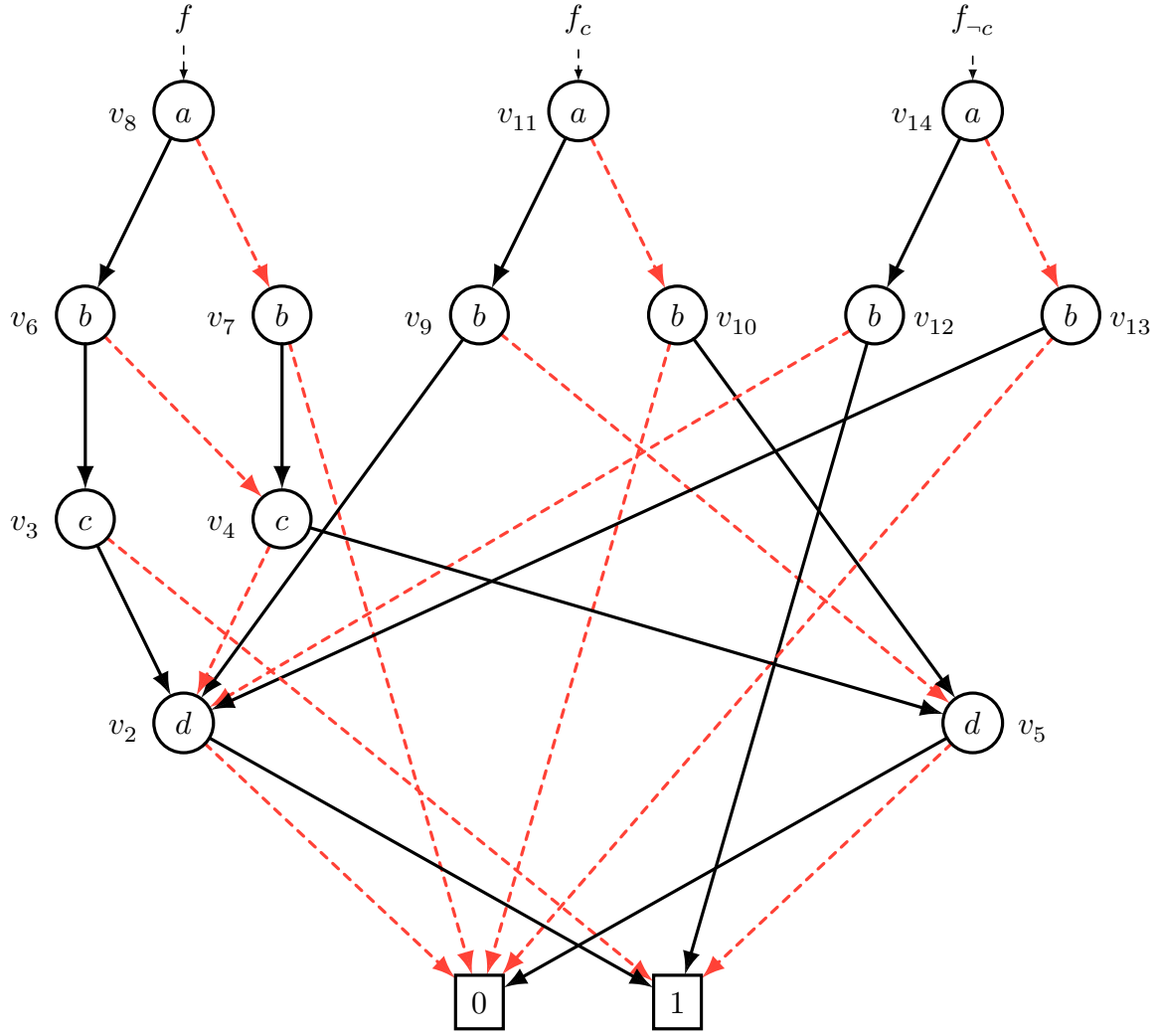
Using procedure `robdd_build` to f_c , we have

$$\begin{aligned} & \text{robdd_build}(abd + (a\neg b + \neg ab)(\neg d), 1) \\ & \xrightarrow{\eta} \text{robdd_build}(bd + \neg b\neg d, 2) \\ & \quad \xrightarrow{\eta} \text{robdd_build}(d, 4) \\ & \quad \quad v_2 = (d, v_1, v_0) \\ & \quad \xrightarrow{\lambda} \text{robdd_build}(\neg d, 4) \\ & \quad \quad v_5 = (d, v_0, v_1) \\ & \quad \quad v_9 = (b, v_2, v_5) \\ & \xrightarrow{\eta} \text{robdd_build}(b\neg d, 2) \\ & \quad \xrightarrow{\eta} \text{robdd_build}(\neg d, 4) \\ & \quad \quad v_5 = (d, v_0, v_1) \\ & \quad \xrightarrow{\lambda} \text{robdd_build}(0, 4) \\ & \quad \quad v_0 \\ & \quad \quad v_{10} = (b, v_5, v_0) \\ & v_{11} = (a, v_9, v_{10}) \end{aligned}$$

Using procedure `robdd_build` to $f_{\neg c}$, we have

$$\begin{aligned} & \text{robdd_build}(ab + (a\neg b + \neg ab)d, 1) \\ & \xrightarrow{\eta} \text{robdd_build}(b + \neg bd, 2) \\ & \quad \xrightarrow{\eta} \text{robdd_build}(1, 4) \\ & \quad \quad v_1 \\ & \quad \xrightarrow{\lambda} \text{robdd_build}(d, 4) \\ & \quad \quad v_2 = (d, v_1, v_0) \\ & \quad \quad v_{12} = (b, v_1, v_2) \\ & \xrightarrow{\lambda} \text{robdd_build}(bd, 2) \\ & \quad \xrightarrow{\eta} \text{robdd_build}(d, 4) \\ & \quad \quad v_2 = (d, v_1, v_0) \\ & \quad \xrightarrow{\lambda} \text{robdd_build}(0, 4) \\ & \quad \quad v_0 \\ & \quad \quad v_{13} = (b, v_2, v_0) \\ & v_{14} = (a, v_{12}, v_{13}) \end{aligned}$$

Hence, the shared ROBDDs of f , f_c and $f_{\neg c}$ are as shown as below (Figure 1).

Figure 1: The shared ROBDDs of f , f_c and f_{-c} .

Red dashed edge represents 0-edge, and black solid edge represents 1-edge.

1. (c)

Since

$$\exists c. f = f_c + f_{-c} = \text{ITE}(f_c, 1, f_{-c}) = \text{ITE}(v_{11}, 1, v_{14})$$

we can use the ROBDD in Figure 1 to compute the ROBDD of $\text{ITE}(f_c, 1, f_{-c})$. Hence we have:

$$\begin{aligned} & \text{ITE}(f_c, 1, f_{-c}) \\ &= \text{ITE}(v_{11}, 1, v_{14}) \\ &= \text{ITE}(a, \text{ITE}(v_9, 1, v_{12}), \text{ITE}(v_{10}, 1, v_{13})) \\ &= \text{ITE}(a, \text{ITE}(b, \text{ITE}(v_2, 1, 1), \text{ITE}(v_5, 1, v_2)), \text{ITE}(b, \text{ITE}(v_5, 1, v_2), \text{ITE}(0, 1, 0))) \end{aligned}$$

since $v_2 = d$ and $v_5 = \neg d$, we have

$$\begin{aligned} \text{ITE}(v_2, 1, 1) &= 1 \\ \text{ITE}(v_5, 1, v_2) &= \text{ITE}(\neg d, 1, d) = 1 \\ \text{ITE}(0, 1, 0) &= \text{ITE}(d, 1, 0) = 0. \end{aligned}$$

Therefore,

$$\begin{aligned}
& \text{ITE}(f_c, 1, f_{-c}) \\
&= \text{ITE}(a, \text{ITE}(b, 1, 1), \text{ITE}(b, 1, 0)) \\
&= \text{ITE}(a, 1, b) \\
&= a + b
\end{aligned}$$

The ROBDD of $\text{ITE}(f_c, 1, f_{-c}) = \text{ITE}(a, 1, b) = a + b$ is as shown as below (Figure 2).

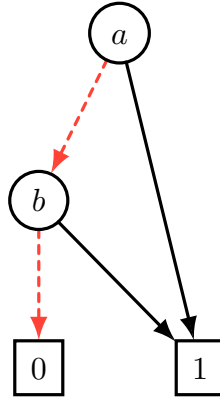


Figure 2: The ROBDD of $\text{ITE}(f_c, 1, f_{-c})$.

2 SAT Solving

3 Combinational Equivalence Checking

4 Characteristic Function

5 Sequential Equivalence Checking